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A two-grid combined finite element-upwind finite volume method for a nonlinear convection-dominated diffusion reaction equation

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ABSTRACT

In this paper, we first present a combined finite element-upwind finite volume method for a fully nonlinear convection-dominated diffusion reaction equation and derive its a priori optimal error estimate in H^1-norm and sub-optimal error estimate in L^2-norm for piecewise linear finite element combining with the first order upwind finite volume scheme. Then we study a type of two-grid method for the nonlinear convection-dominated transport equation together with the combined finite element-upwind finite volume method on the fine grid T_h and the streamline diffusion finite element scheme on the coarse grid T_H, which not only significantly reduces the computational cost on nonlinear iterations but also remains the numerical computation stabilized and the approximation accuracy unchanged. A priori error estimate of such designed two-grid method in H^1-norm is proved to be $O(h + H^{\frac{3}{2}})$, showing that the two-grid method achieves the optimal approximation as long as the mesh sizes satisfy $h = O(H^{\frac{3}{2}})$. Finally, a numerical example is carried out to verify the accuracy and efficiency of the present numerical method.

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1. Introduction

We consider the following general fully nonlinear convection-dominated diffusion reaction equation

$$\begin{cases} -\nabla \cdot (a(u)\nabla u) + \nabla \cdot (\vec{b}(u)u) = f(u), & x \in \Omega \\ u = 0. & x \in \Gamma \end{cases} \tag{1}$$

in a domain $\Omega \subset R^n$ with boundary Γ. We make the following assumptions for problem (1). The coefficient functions $a(u), \vec{b}(u) = (b_1(u), \ldots, b_n(u))^T, f(u)$ are all bounded and Lipschitz continuous with respect to u, and there exist positive constants A_0, C such that $0 < A_0 \le a(u) < C$, $a(u), b_i(u), f(u) \in C_b^2(\Omega),\ i = 1, 2, \ldots, n$, where $C_b^2(\Omega)$ is a class of twice continuously differentiable functions in Ω such that all derivatives of the functions up to and including second order are bounded in Ω. In addition, $\nabla \cdot b(u) > 0$, and $|\vec{b}(u)| \gg a(u)$ which reflects the dominant convection effect.

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The investigation of the convection–diffusion problems has been the object of an increasing interest of a number of specialists as well as mathematicians. Special attention is paid particularly to problem with convection dominating over diffusion. It is well known that the straightforward application of the finite element method to singularly perturbed convection–diffusion problems may lead to spurious oscillations in the approximate solution [1]. The reason for this drawback is that the discrete method loses most of the stability properties of the continuous problem. Several methods have been intensively studied to remove such a drawback. A popular idea is to add stabilization terms to the formulation of the problem, such as the Streamline-Upwind/Petrov–Galerkin (SUPG) method [2,3], or streamline diffusion method [4–8], or Galerkin-least-squares method [9–13], however, these schemes need to manually tune the stabilization parameters in order to achieve a good stability. Although there exists some formula to help to find an appropriate stabilization parameter for linear PDE, it is difficult to define such formula for a nonlinear problem. In this sense, upwind finite volume scheme seems more promising because its upwind parameter can be automatically obtained based on the numerical flux.

On the other hand, in many physical problems the conservation of some quantities such as mass or momentum is a basic property of the mathematical model. Hence, it is important to study proper discretization to preserve the same feature, which is not generally guaranteed by the stabilized finite element methods. Upwinding and conservation appear naturally when we are dealing with a finite volume technique.

Based on an upwind modification of a standard Galerkin discretization, the problem of finding a finite element scheme for a convection–diffusion problem whose solution satisfies a discrete mass conservation law was first addressed in [14]. Feistauer [15] proposed and analyzed a semi-implicit scheme for a nonlinear convection–diffusion problem based on a node-centered finite volume approach which prove the convergence of the discrete solution to the exact solution. To maintain the superiority of finite element method on the convergence accuracy and the convenience of dealing with the natural boundary conditions, at the same time, overcome the dominated convection effect and stabilize the numerical computation, in this paper we present a combined finite element-upwind finite volume method [14–18] to solve a fully nonlinear convection-dominated diffusion reaction equation, which discretizes the diffusion and reaction terms with standard finite element scheme. Hashim and Hu [19] applied a similar approach to the semi-linear convection–diffusion equation with a simpler convection term $b(u) \cdot \nabla u$, but none of them uses this method for a fully nonlinear convection-dominated diffusion reaction equation, in which the convection term is in divergence form $\nabla \cdot (b(u)u)$, which is the most general case of convection effect, and the reaction term $f(u)$ is nonlinear too. In this paper we design an appropriate combined finite element-upwind finite volume scheme to solve the aforementioned nonlinear PDE, analyze its a priori error estimate, and obtain the optimal error estimate in H^1 norm and sub-optimal error estimate in L^2 norm for a piecewise linear finite element combining with the first order upwind finite volume scheme, where, to our knowledge, L^2 norm error estimate is established for the present nonlinear model for the first time in this paper.

Besides the stability issue induced by the dominant convection effect, the nonlinear iteration is another important issue when we carry out a numerical simulation for a nonlinear problem on a fine grid, which is always time-consuming and thus expensive. It is well known that two-grid method [20,21] is a discretization technique for efficiently solving nonlinear equations based on two grids with different sizes. This method iteratively approximates the solution of the original nonlinear problem on a coarse grid first, then the symmetry and linear part of the equation are modified on a fine grid. Since the two-grid method only conducts the nonlinear iteration on the coarse grid, and solve a linear equation on the fine grid, the entire computational cost is thus significantly reduced. Two-grid method has been successfully used for solving elliptic boundary value problems [20,21], Stokes equation [22], the steady-state Navier–Stokes equations [23] and other partial differential equations. Wu [24] applied two-grid method combining with mixed finite element method to the reaction–diffusion equations. Chen [25] constructed a two-grid method for the expanded mixed finite element solution of a semilinear reaction–diffusion equations. However, to our knowledge, no one has ever used the two-grid approach in the combined finite element-upwind finite volume method yet.

In this paper, based on two conforming linear finite element spaces V_H defined on the coarse grid with grid size H and V_h defined on the fine grid with grid size h, we consider a two-grid combined finite element-upwind finite volume method for the fully nonlinear convection-dominated diffusion reaction Eq. (1). With the proposed techniques, we solve the original nonlinear problem in the coarse finite element space with a stabilized streamline diffusion-finite element discretization and Picard's linearization, then solve the corresponding linearized problem in the fine finite element space with the combined finite element-upwind finite volume method. Thus, based on the existing convergence result of streamline diffusion finite element scheme [4–8] and the newly obtained convergence theorem of the combined finite element-upwind finite volume method, we are able to attain an optimal error estimate for the present two-grid method. A remarkable fact about this approach is that the coarse grid can be quite coarse and thus the work for solving the nonlinear problem is relatively negligible, in the meanwhile, the approximation accuracy remains the same with that of the numerical method which is adopted on the fine grid.

The rest of this paper is organized as follows. In Section 2, we describe the combined finite element-upwind finite volume method and conduct its a priori error estimate for the nonlinear problem (1). The two-grid combined finite element-upwind finite volume method and its a priori error estimate are given in Section 3. A numerical example is presented to verify the theoretical results in Section 4.

Throughout this paper, the letter C or with its subscript denotes a generic positive constant which does not depend on the mesh sizes and may represent different value at different occurrences.

2. The combined finite element-upwind finite volume method

2.1. The finite element space and dual mesh

Let $L^p(\Omega)$ denotes the linear space of measurable pth power integrable function on Ω endowed with norm $\|\cdot\|_{0,p,\Omega}$, and $W_p^m(\Omega)$ the Sobolev space of function which together with their derivatives up to m are in $L^p(\Omega)$. The norm in $W_p^m(\Omega)$ is denoted by

$$\|u\|_{s,p,\Omega} = \left(\int_\Omega \sum_{|\alpha|\le s} |D^\alpha u|^p dx\right)^{1/p}, \tag{2}$$

and the seminorm

$$|u|_{m,p,\Omega} = \left(\int_\Omega \sum_{|\alpha|=m} |D^\alpha u|^p dx\right)^{1/p}, \tag{3}$$

with the standard modification for $p = \infty$. In order to simplify the notation, we denote $W_2^s(\Omega)$ by $H^s(\Omega)$ and omit the index $p = 2$ and Ω whenever possible, that is, $\|u\|_{s,2,\Omega} = \|u\|_{s,2} = \|u\|_s$. We denote by $H_0^1(\Omega)$ the subspace of $H^1(\Omega)$ of functions vanishing on the boundary Γ.

The weak form of (1) is to find $u \in H^2(\Omega) \cap H_0^1(\Omega)$ such that

$$(a(u)\nabla u, \nabla v) + (\nabla \cdot (\vec{b}(u)u), v) = (f(u), v) \quad \forall v \in H_0^1(\Omega). \tag{4}$$

Let us consider a regular triangulation $T_h = \{e\}$ defined over $\bar{\Omega}$ where each element of T_h is a closed triangle. We denote the internal vertices by $p_i (1 \le i \le N)$ and the boundary vertices on Γ by $p_i (N + 1 \le i \le K)$, where N and K are positive integers dependent on the triangulation.

Let φ_i, $1 \le i \le K$, be the continuous function in $\bar{\Omega}$ such that $\varphi_i(p)$ is linear on each $e \in T_h$ and $\varphi_i(p_j) = \delta_{i,j}$ for all p_j. We denoted by V_h the linear span of φ_i, $1 \le i \le K$, and by V_{0h} the linear span of φ_i, $1 \le i \le N$, such that

$$\begin{aligned} V_h &= \{\varphi : \varphi \in C(\bar{\Omega}), \varphi \text{ is linear function on } e, \forall e \in T_h; \varphi = 0 \text{ on } \Gamma\}, \\ V_{0h} &= \{\varphi : \varphi \in V_h, \varphi(p_k) = 0,\ k = N + 1, \ldots, K\}. \end{aligned} \tag{5}$$

We define h_e as the maximum side length of e, κ_e as the minimum perpendicular length of e, and let $h = \max_{e\in T_h}\{h_e\}$, $\kappa = \min_{e\in T_h}\{\kappa_e\}$, $\frac{h}{\kappa} \le C$ to make sure each element is shape regular.

We consider the dual decomposition $\hat{T}_h = \{\Omega_i\}$, as shown in Fig. 1, where Ω_i is the circumcentric domain associated with nodal point p_i

$$\Omega_i = \bigcup_e \{\Omega_i^e, e \in T_h \text{ such that } p_i \text{ is a vertex of } e\}, \tag{6}$$

where

$$\Omega_i^e = \{p : p \in e, |p - p_i| \le |p - p_j|,\ \forall p_j \in e\}, \tag{7}$$

we define the index set

$$\Lambda_i = \{j \ne i; p_j \text{ is adjacent to } p_i\} \tag{8}$$

and

$$\Lambda = \{j \in \wedge_i, e \in T_h\}. \tag{9}$$

We denote the boundary of Ω_i by Γ_i and $\Gamma_{i,j} = \Gamma_i \cap \Gamma_j$, $i = 1, 2 \ldots, K$, $j \in \wedge_i$, $T_h(i) = \{e \in T_h \text{ such that } p_i \in e\}$, $T_h(ij) = T_h(i) \cap T_h(j)$ and n_{ij} is the outer normal of $\Gamma_{i,j}$ in Ω_i. By making use of the characteristic function μ_i of circumcentric domain Ω_i, the mass lumping operator $\wedge$ is defined by, $\wedge : w \in C(\bar{\Omega}) \to \hat{w} \in L^\infty(\Omega)$, such that [26,27]

$$\hat{w}(p) = \sum_{i=1}^K \omega(p_i)\mu_i(p), \tag{10}$$

where $\hat{w}(p) = w(p_i) = w_i$, $\forall p \in \Omega_i$.

We state here some fundamental lemmas for later use. Their proofs can be found in [26,27].

Lemma 2.1. *For all $w \in V_h$*

$$(\nabla w, \nabla \varphi_i) = -\sum_{j\in\wedge_i} (w_j - w_i)|\Gamma_{ij}|/|p_i - p_j|, \quad 1 \le i \le K, \tag{11}$$

where φ_i is the base function of finite element space V_h.

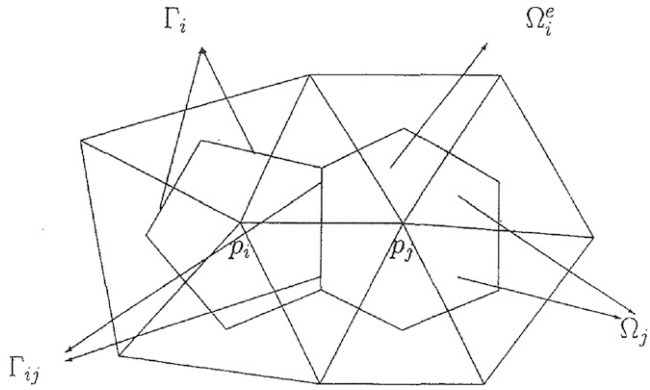

Fig. 1. The circumcentric domain.

Lemma 2.2. *For all $w \in V_h$ with $p \geq 1$*

$$\|\hat{w} - w\|_{0,p,\Omega} \leq ch|w|_{1,p,\Omega}. \tag{12}$$

2.2. The combined finite element-upwind finite volume scheme

The combined finite element-upwind finite volume scheme of Eq. (1) is to find $u_h \in V_h \subset H^2(\Omega) \cap H_0^1(\Omega)$ such that

$$(a(u_h)\nabla u_h, \nabla v_h) + B_h(\vec{b}(u_h), u_h, v_h) = (f(u_h), v_h), \quad \forall v_h \in V_h. \tag{13}$$

In the following we derive the $B_h(\vec{b}(u_h), u_h, v_h)$ by introducing its continuous form first as follows

$$\begin{aligned} B(\vec{b}(u), u, v) &:= (\nabla \cdot (\vec{b}(u)u), v) \\ &\approx \sum_{i=1}^{K} \int_{\Omega_i} (\nabla \cdot (\vec{b}(u)u))\hat{v}dx \\ &= \sum_{i=1}^{K} v(p_i) \int_{\Omega_i} (\nabla \cdot (\vec{b}(u)u))dx \\ &= \sum_{i=1}^{K} v(p_i) \sum_{j \in \wedge_i} \int_{\Gamma_{ij}} (\vec{b}(u) \cdot n_{ij})uds, \end{aligned} \tag{14}$$

where $\hat{v}$ is the mass lumping function, defined as (10). Replacing the function u on Γ_{ij} by some convex combination of the nodal values of $u(p_i)$ and $u(p_j)$ with parameter r_{ij}, we can define

$$B_h(\vec{b}(u_h), u_h, v_h) := \sum_{i=1}^{K} v_h(p_i) \sum_{j \in \wedge_i} \left(r_{ij}u_h(p_i) + (1 - r_{ij})u_h(p_j)\right) \beta_{ij}, \tag{15}$$

where $\beta_{ij} = \int_{\Gamma_{ij}} \vec{b}(u_h) \cdot n_{ij}ds$,

$$r_{ij} = \begin{cases} 1 & \text{if } \beta_{ij} > 0 \\ 0.5 & \text{if } \beta_{ij} = 0 \\ 0 & \text{if } \beta_{ij} < 0. \end{cases} \tag{16}$$

2.3. The H^1-norm error estimate

Subtracting (13) from (4), we can obtain the error equation as follows

$$(a(u)\nabla u - a(u_h)\nabla u_h, \nabla v_h) + B(\vec{b}(u), u, v_h) - B_h(\vec{b}(u_h), u_h, v_h) = (f(u) - f(u_h), v_h). \tag{17}$$

We set $u - u_h = (u - I_h u) + (I_h u - u_h) = \eta + \xi$, where I_h is the interpolation operator, it is well known that the following interpolation approximation error estimates hold [28]:

Lemma 2.3.

$$\|\eta\|_0 \leq Ch^2\|u\|_2, \qquad \|\eta\|_1 \leq Ch\|u\|_2. \tag{18}$$

Let $v_h = \xi$, the error Eq. (17) can be written as

$$(a(u_h)\nabla\xi, \nabla\xi) + (a(u_h)\nabla\eta, \nabla\xi) + (a(u) - a(u_h))(\nabla u, \nabla\xi) + B(\vec{b}(u), u, \xi) - B_h(\vec{b}(u_h), u_h, \xi) = (f(u) - f(u_h), \xi). \tag{19}$$

Since $a(u)$ and $f(u)$ are Lipschitz continuous functions, $a(u) \geq A_0 > 0$, and we assume $u \in W^{1,\infty}$, then by using ε-inequality $pq \leq \varepsilon p^2 + \frac{1}{4\varepsilon}q^2$, we have

$$\begin{aligned}
&(a(u_h)\nabla\xi, \nabla\xi) \geq A_0\|\nabla\xi\|_0^2,\\
&(a(u_h)\nabla\eta, \nabla\xi) \leq C\|\nabla\eta\|_0\|\nabla\xi\|_0,\\
&(a(u) - a(u_h))(\nabla u, \nabla\xi) \leq C\|\eta + \xi\|_0\|\nabla\xi\|_0 \leq C(\|\xi\|_0^2 + \|\eta\|_0^2 + \varepsilon\|\nabla\xi\|_0^2),\\
&(f(u) - f(u_h), \xi) \leq C\|\eta + \xi\|_0\|\xi\|_0 \leq C(\|\xi\|_0^2 + \|\eta\|_0^2 + \varepsilon\|\xi\|_0^2),
\end{aligned} \tag{20}$$

thus for a sufficiently small $\varepsilon > 0$,

$$A_0\|\nabla\xi\|_0^2 \leq Ch\|\nabla\xi\|_0 + Ch^2 + \|\xi\|_0^2 + |B(\vec{b}(u), u, \xi) - B_h(\vec{b}(u_h), u_h, \xi)|, \tag{21}$$

where, Poincare inequality is applied. By triangle inequality, we get

$$\begin{aligned}
|B(\vec{b}(u), u, \xi) - B_h(\vec{b}(u_h), u_h, \xi)| \;\leq\; & |B(\vec{b}(u), u, \xi) - B(\vec{b}(u_h), u, \xi)| + |B(\vec{b}(u_h), u, \xi) - B(\vec{b}(u_h), u_h, \xi)|\\
&+ |B(\vec{b}(u_h), u_h, \xi) - B(\vec{b}(u_h), u_h, \hat{\xi})|\\
&+ |B(\vec{b}(u_h), u_h, \hat{\xi}) - B_h(\vec{b}(u_h), u_h, \xi)|\\
:= \; & I_1 + I_2 + I_3 + I_4,
\end{aligned} \tag{22}$$

where

$$\begin{aligned}
I_1 &= |(\nabla\cdot(\vec{b}(u)u), \xi) - (\nabla\cdot(\vec{b}(u_h)u), \xi)|\\
&= |((\vec{b}(u) - \vec{b}(u_h))u, \nabla\xi)|\\
&\leq C\|\eta + \xi\|_0\|\nabla\xi\|_0\\
&\leq C(\|\xi\|_0^2 + \|\eta\|_0^2 + \varepsilon\|\nabla\xi\|_0^2),
\end{aligned} \tag{23}$$

$$\begin{aligned}
I_2 &= |(\nabla\cdot(\vec{b}(u_h)u), \xi) - (\nabla\cdot(\vec{b}(u_h)u_h), \xi)|\\
&= |(\vec{b}(u_h)(u - u_h), \nabla\xi)|\\
&\leq C\|\eta + \xi\|_0\|\nabla\xi\|_0\\
&\leq C(\|\xi\|_0^2 + \|\eta\|_0^2 + \varepsilon\|\nabla\xi\|_0^2),
\end{aligned} \tag{24}$$

$$\begin{aligned}
I_3 &= |(\nabla\cdot(\vec{b}(u_h)u_h), \xi) - (\nabla\cdot(\vec{b}(u_h)u_h), \hat{\xi})|\\
&= (\vec{b}'(u_h)\cdot\nabla u_h u_h + \vec{b}(u_h)\cdot\nabla u_h, \xi - \hat{\xi})\\
&= \left(\frac{1}{2}\vec{b}'(u_h)\cdot\nabla u_h^2 + \vec{b}(u_h)\cdot\nabla u_h, \xi - \hat{\xi}\right)\\
&\leq C(\|\nabla u_h^2\|_0 + \|\nabla u_h\|_0)\|\xi - \hat{\xi}\|_0\\
&\leq C(\|\nabla\eta + \nabla\xi\|_0^2 + \|\nabla u\|_0^2 + \|\nabla\eta + \nabla\xi\|_0 + \|\nabla u\|_0)h\|\nabla\xi\|_0\\
&\leq Ch(\|\nabla\xi\|_0 + \|\nabla\xi\|_0^2 + \|\nabla\xi\|_0^3),
\end{aligned} \tag{25}$$

where we apply Lemma 2.2 in (25).

$$\begin{aligned}
I_4 &= \left|\sum_{i=1}^{K}\xi(p_i)\sum_{j\in\wedge_i}\int_{\Gamma_{ij}} u_h\vec{b}(u_h)\cdot n_{ij}ds - \sum_{i=1}^{K}\xi(p_i)\sum_{j\in\wedge_i}\int_{\Gamma_{ij}} u_h^{ij}\vec{b}(u_h)\cdot n_{ij}ds\right|\\
&= \left|\sum_{i=1}^{K}\xi(p_i)\sum_{j\in\wedge_i}\int_{\Gamma_{ij}} (u_h - u_h^{ij})\vec{b}(u)\cdot n_{ij}ds\right|,
\end{aligned} \tag{26}$$

where $u_h^{ij} = r_{ij}u_h(p_i) + (1 - r_{ij})u_h(p_j)$, then from Lemmas 2.1 and 2.2, using the error estimate in L^∞ norm and inverse inequality, we get

$$\begin{aligned}
I_4 &= \left|\sum_{e\in T_h}\sum_{p_i,p_j\in e, i<j}(\xi(p_i) - \xi(p_j))\int_{\Gamma_{ij}^e}(u_h - u_h^{ij})\vec{b}(u)\cdot n_{ij}ds\right|\\
&\leq C\sum_{e\in T_h} h_e\|\nabla\xi\|_{0,\infty}\sum_{p_i,p_j\in e, i<j}|\Gamma_{ij}^e|\,\|\nabla\cdot\vec{b}(u_h)\|_{0,\infty}\|u_h - h_h^{ij}\|_{0,\infty}
\end{aligned}$$

$$
\begin{aligned}
&\le C\sum_{e\in T_h} h_e\|\nabla\xi\|_{0,\infty}h_e^{n-1}h\|\nabla u_h\|_{0,\infty}\\
&\le C\sum_{e\in T_h} h_e^{n+1}h_e^{-\frac{n}{2}}\|\nabla\xi\|_0 h_e^{-\frac{n}{2}}\|\nabla u_h\|_0\\
&\le Ch\|\nabla(u_h-u+u)\|_0\|\nabla\xi\|_0\\
&\le Ch(\|\nabla\eta\|_0^2+\|\nabla\xi\|_0^2+\|\nabla\xi\|_0).
\end{aligned}\tag{27}
$$

Therefore, by Lemma 2.3 we have

$$
|B(\vec{b}(u),u,\xi)-B_h(\vec{b}(u_h),u_h,\xi)|\le Ch^2+C\|\xi\|_0^2+\varepsilon\|\nabla\xi\|_0^2+Ch\|\nabla\xi\|_0^3,\tag{28}
$$

then (21) leads to

$$
\|\xi\|_1\le C(h+\|\xi\|_0+h\|\xi\|_1^2)\le C(h+\|\xi\|_0).\tag{29}
$$

Combining with Lemma 2.3, we obtain

$$
\|u-u_h\|_1\le C(h+\|u-u_h\|_0).\tag{30}
$$

In the next section we will estimate $\|u-u_h\|_0$, then the error estimates in both H^1 and L^2 norms will be attained.

2.4. The L^2-norm error estimate

To analyze the L^2 error estimate, suppose $a(u), b_i(u), f(u)\in C^2(\Omega)$, $u\in W^1_\infty(\Omega)$, we define the following linearized adjoint problem, let $e=u-u_h$, find $\phi\in H^2(\Omega)\cap H_0^1(\Omega)$ such that

$$
\begin{cases}
-\nabla\cdot(a(u)\nabla\phi)+a'(u)\nabla u\cdot\nabla\phi-\vec{b}(u)\cdot\nabla\phi-\vec{b}'(u)u\cdot\nabla\phi-f'(u)\phi=e & u\in\Omega\\
\phi=0 & u\in\Gamma,
\end{cases}\tag{31}
$$

where u is the exact solution of (1). Our assumptions on $a(u), \vec{b}(u), f(u)$ and u imply that there is a unique solution ϕ to the problem (31) and ϕ satisfies the following elliptic regularity [29,30]

$$
\|\phi\|_2\le C\|e\|_0.\tag{32}
$$

In fact

$$
\|e\|_0^2=(e,-\nabla\cdot(a(u)\nabla\phi))+(e,a'(u)\nabla u\cdot\nabla\phi)-(e,\vec{b}(u)\cdot\nabla\phi)-(e,\vec{b}'(u)u\cdot\nabla\phi)-(e,f'(u)\phi).\tag{33}
$$

For the first term and third term on the right hand side of (33), we have the following estimate

$$
\begin{aligned}
(e,-\nabla\cdot(a(u)\nabla\phi)) &= (a(u_h)\nabla e,\nabla\phi)+((a(u)-a(u_h))\nabla e,\nabla\phi)\\
&= (a(u_h)\nabla e,\nabla(\phi-\phi_h))+(a(u_h)\nabla e,\nabla\phi_h)+((a(u)-a(u_h))\nabla e,\nabla\phi)\\
&= (a(u_h)\nabla e,\nabla(\phi-\phi_h))+((a(u_h)-a(u)+a(u))\nabla u,\nabla\phi_h)\\
&\quad -(a(u_h)\nabla u_h,\nabla\phi_h)+((a(u)-a(u_h))\nabla e,\nabla\phi)\\
&= (a(u_h)\nabla e,\nabla(\phi-\phi_h))+((a(u_h)-a(u))\nabla u,\nabla\phi_h)+((a(u)-a(u_h))\nabla e,\nabla\phi)\\
&\quad +(a(u)\nabla u,\nabla\phi_h)-(a(u_h)\nabla u_h,\nabla\phi_h),
\end{aligned}\tag{34}
$$

$$
\begin{aligned}
-(e,\vec{b}(u)\cdot\nabla\phi) &= (-\vec{b}(u_h)e,\nabla\phi)-((\vec{b}(u)-\vec{b}(u_h))e,\nabla\phi)\\
&= (-\vec{b}(u_h)e,\nabla(\phi-\phi_h))-(\vec{b}(u_h)e,\nabla\phi_h)-((\vec{b}(u)-\vec{b}(u_h))e,\nabla\phi)\\
&= (-\vec{b}(u_h)e,\nabla(\phi-\phi_h))+((\vec{b}(u)-\vec{b}(u_h))u,\nabla\phi_h)+((\vec{b}(u_h)-\vec{b}(u))e,\nabla\phi)\\
&\quad +(\nabla\cdot(\vec{b}(u)u),\phi_h)-(\nabla\cdot(\vec{b}(u_h)u_h),\phi_h),
\end{aligned}\tag{35}
$$

by (17), then (33) can be further induced as

$$
\begin{aligned}
\|e\|_0^2 &= (a(u_h)\nabla e,\nabla(\phi-\phi_h))+((a(u_h)-a(u))\nabla u,\nabla\phi_h)+((a(u)-a(u_h))\nabla e,\nabla\phi)\\
&\quad +(e,a'(u)\nabla u\cdot\nabla\phi)-(\vec{b}(u_h)e,\nabla(\phi-\phi_h))+(\vec{b}(u)-\vec{b}(u_h))(u,\nabla\phi_h)\\
&\quad -(\vec{b}(u)-\vec{b}(u_h))(e,\nabla\phi)-(\vec{b}'(u)e,u\nabla\phi)-B(\vec{b}(u_h),u_h,\phi_h)\\
&\quad +B_h(\vec{b}(u_h),u_h,\phi_h)+(f(u)-f(u_h),\phi_h)-f'(u)(e,\phi)\\
&= (a(u_h)\nabla e,\nabla(\phi-\phi_h))+(a'(\zeta)e\nabla u,\nabla(\phi-\phi_h))+(a''(\varsigma)e^2\nabla u,\nabla\phi)\\
&\quad +\left(\frac{1}{2}a'(\zeta)\nabla e^2,\nabla\phi\right)-(\vec{b}(u_h)e,\nabla(\phi-\phi_h))-(\vec{b}'(\zeta)eu,\nabla(\phi-\phi_h))\\
&\quad +(\vec{b}''(\varsigma)e^2u,\nabla\phi)+\left(\frac{1}{2}\vec{b}'(\zeta)e^2,\nabla\phi\right)-B(\vec{b}(u_h),u_h,\phi_h)\\
&\quad +B_h(\vec{b}(u_h),u_h,\phi_h)+(f'(u)e,\phi_h-\phi)+(f''(\varsigma)e^2,\phi_h).
\end{aligned}\tag{36}
$$

Using the same estimate as (25)–(27), we obtain

$$\begin{aligned}|B(\vec{b}(u_h), u_h, \phi_h) - B_h(\vec{b}(u_h), u_h, \phi_h)| &\le |B(\vec{b}(u_h), u_h, \phi_h) - B(\vec{b}(u_h), u_h, \hat{\phi}_h)| \\ &\quad + |B(\vec{b}(u_h), u_h, \hat{\phi}_h) - B_h(\vec{b}(u_h), u_h, \phi_h)| \\ &\le Ch(\|\nabla e\|_0^2 + \|\nabla e\|_0)\|\nabla \phi_h\|_0 + Ch\|\nabla \phi_h\|_0, \end{aligned} \tag{37}$$

then

$$\|e\|_0^2 \le Ch(\|e\|_1 + \|e\|_1^2)\|\phi\|_2 + C(\|e\|_0^2 + \|\nabla e\|_0^2)\|\nabla \phi_h\|_0 + Ch(\|\nabla e\|_0^2 + \|\nabla e\|_0)\|\nabla \phi_h\|_0 + Ch\|\nabla \phi_h\|_0, \tag{38}$$

by (30), we have

$$\|e\|_1 \le C(h + \|e\|_0), \tag{39}$$

combining with (32), then

$$\|e\|_0 \le C(h + \|e\|_0^2). \tag{40}$$

By the compactness argument [31,32], we know that $\|e\|_0 \to 0$, then $\|e\|_0^2 \to 0$ quadratically as $h \to 0$ in contrast to $\|e\|_0 \to 0$. Therefore,

$$\|e\|_0 \le Ch. \tag{41}$$

Combining (39) and (41), we have the following convergence theorem.

Theorem 2.1. *Suppose u is the real solution of Eq. (1) and u_h the numerical solution of (13), $u \in H^2(\Omega) \cap H_0^1(\Omega) \cap W^{1,\infty}(\Omega)$, then the combined finite element-upwind finite volume scheme (13) holds the following error estimates to approximate (1):*

$$\|u - u_h\|_0 + \|u - u_h\|_1 \le Ch. \tag{42}$$

3. The two-grid method

In this section, we present a type of two-grid method to efficiently solve the fully nonlinear convection-dominated transport problem (1) on the basis of two different finite element spaces. The idea of this two-grid method is to solve the original nonlinear problem on a coarse grid with streamline diffusion finite element method, and then solve a corresponding linear problem on the fine grid with the combined finite element-upwind finite volume method. First of all, we introduce two quasi-uniform triangulations of Ω, T_H, and T_h with two different mesh sizes H and h ($H \gg h$), then define their corresponding piecewise linear finite spaces V_H and V_h satisfying $V_H \subset V_h$.

It is well known that the streamline diffusion finite element method introduced by Hughes and Brooks [2,3] holds the following global error estimate in L^2 norm if the diffusion coefficient is less than mesh size h and piecewise polynomials of degree k are used [4]:

$$\|u - u_h\|_0 \le Ch^{k+1/2}, \tag{43}$$

which is half order lower than the optimal one but still half order higher than the H^1-norm error estimate of the standard Galerkin method. Therefore, we are able to design the following two-grid algorithm and still achieve an efficient nonlinear iteration on the coarse grid only without losing the accuracy in H^1 norm, as shown in Theorem 3.1.

Algorithm 3.1. Step 1. Employing the streamline diffusion finite element method, on the coarse grid T_H, find $u_H \in V_H$, such that

$$\begin{aligned}&(a(u_H)\nabla u_H, \nabla v_H) + \delta(H)(a(u_H)\nabla u_H, \nabla \cdot \vec{b}(u_H)\nabla v_H) \\ &\quad - (\vec{b}(u_H)u_H, \nabla v_H) + \delta(H)(\nabla \cdot \vec{b}(u_H)u_H, \vec{b}(u_H) \cdot \nabla v_H) + \delta(H)(\vec{b}(u_H)\nabla u_H, \vec{b}(u_H) \cdot \nabla v_H) \\ &= (f(u_H), v_H + \delta(H)\vec{b}(u_H) \cdot \nabla v_H), \quad \forall v_H \in V_H. \end{aligned} \tag{44}$$

Step 2. Employing an upwind technique, on the fine grid T_h, find $u_h^* \in V_h$, such that

$$(a(u_H)\nabla u_h^*, \nabla v_h) + B_h(\vec{b}_H, u_h^*, v_h) = (f(u_H), v_h), \quad \forall v_h \in V_h. \tag{45}$$

Now we consider the error estimates in H_1-norm for the two-grid combined finite element-upwind finite volume method Algorithm 3.1.

Theorem 3.1. *Assume that u_h and u_h^* are the solutions obtained by (13) and Algorithm 3.1 for $H \ll 1$, then*

$$\|u_h - u_h^*\|_1 \le C\left(h + H^{\frac{3}{2}}\right); \tag{46}$$

$$\|u - u_h^*\|_1 \le C\left(h + H^{\frac{3}{2}}\right), \tag{47}$$

where u is the real solution of Eq. (1).

Proof. Subtract (45) from (13), we get

$$(a(u_h)\nabla u_h - a(u_H)\nabla u_h^*, \nabla v_h) + B_h(\vec{b}(u_h), u_h, v_h) - B_h(\vec{b}(u_H), u_h^*, v_h) = (f(u_H) - f(u), v_h). \tag{48}$$

For the first term in Eq. (48), we have

$$\begin{aligned}(a(u_h)\nabla u_h - a(u_H)\nabla u_h^*, \nabla v_h) &= (a(u_h)\nabla u_h - a(u_H)\nabla u_h, \nabla v_h) + (a(u_H)\nabla u_h - a(u_H)\nabla u_h^*, \nabla v_h)\\ &= A_1 + A_2.\end{aligned} \tag{49}$$

Using the Cauchy–Schwarz inequality, we rewrite and estimate A_1 as follows.

$$\begin{aligned}A_1 &= (a(u_h)\nabla(u_h - u), \nabla v_h) + ((a(u_h) - a(u_H))\nabla u, \nabla v_h) + (a(u_H)\nabla(u - u_h), \nabla v_h)\\ &\le C\|\nabla(u_h - u)\|_0\|\nabla v_h\|_0 + C\|u_h - u_H\|_0\|\nabla u\|_\infty\|\nabla v_h\|_0 + C\|\nabla(u_h - u)\|_0\|\nabla v_h\|_0\\ &\le C\|\nabla(u_h - u)\|_0\|\nabla v_h\|_0 + C\|u_h - u + u - u_H\|_0\|\nabla v_h\|_0\\ &\le C\left(h + H^{\frac{3}{2}}\right)\|\nabla v_h\|_0.\end{aligned} \tag{50}$$

For the second and third terms in Eq. (48), we have

$$B_h(\vec{b}(u_h), u_h, v_h) - B_h(\vec{b}(u_H), u_h^*, v_h) = B_1 + B_2 + B_3 + B_4, \tag{51}$$

where

$$\begin{aligned}B_1 &:= B_h(\vec{b}(u_h), u_h, v_h) - B(\vec{b}(u_h), u_h, v_h),\\ B_2 &:= B(\vec{b}(u_h), u_h, v_h) - B(\vec{b}(u_H), u_h, v_h),\\ B_3 &:= B(\vec{b}(u_H), u_h, v_h) - B(\vec{b}(u_H), u_h^*, v_h),\\ B_4 &:= B(\vec{b}(u_H), u_h^*, v_h) - B_h(\vec{b}(u_H), u_h^*, v_h).\end{aligned}$$

Using the same method as in Eqs. (25) and (26), we estimate B_1 and B_4 as

$$\begin{aligned}|B_1| &= |B_h(\vec{b}(u_h), u_h, v_h) - B(\vec{b}(u_h), u_h, \hat{v_h})| + |B(\vec{b}(u_h), u_h, \hat{v_h}) - B(\vec{b}(u_h), u_h, v_h)|\\ &\le Ch(h + \|\nabla v_h\|_0 + \|\nabla v_h\|_0^2) + Ch(\|\nabla v_h\|_0 + \|\nabla v_h\|_0^2 + \|\nabla v_h\|_0^3),\end{aligned} \tag{52}$$

$$\begin{aligned}|B_4| &= |B_h(\vec{b}(u_H), u_h^*, v_h) - B(\vec{b}(u_H), u_h^*, \hat{v_h})| + |B(\vec{b}(u_H), u_h^*, \hat{v_h}) - B(\vec{b}(u_H), u_h^*, v_h)|\\ &\le Ch(h + \|\nabla v_h\|_0 + \|\nabla v_h\|_0^2) + Ch(\|\nabla(u_h - u_h^*)\|_0 + h + C)\|\nabla v_h\|_0.\end{aligned} \tag{53}$$

By the definition of $B(\vec{b}(u_h), u_h, v_h)$, the estimation of B_2 is obtained as

$$\begin{aligned}|B_2| &= |((\vec{b}(u_H)u_h - \vec{b}(u_h)u_h), \nabla v_h)|\\ &= |(\vec{b}(u_H)(u_h - u), \nabla v_h)| + |((\vec{b}(u_H) - \vec{b}(u_h))u, \nabla v_h)| + |(\vec{b}(u_h)(u - u_h), \nabla v_h)|\\ &\le C\|u - u_h\|_0\|\nabla v_h\|_0 + C\|u_H - u_h\|_0\|u\|_\infty\|\nabla v_h\|_0 + C\|u - u_h\|_0\|\nabla v_h\|_0\\ &\le C\left(h + H^{\frac{3}{2}}\right)\|\nabla v_h\|_0.\end{aligned} \tag{54}$$

Let $v_h = u_h - u_h^*$, we have

$$\begin{aligned}B_3 &= -(\vec{b}(u_h)(u_h - u_h^*), \nabla(u_h - u_h^*))\\ &= -\left(\vec{b}(u_H) - \vec{b}(u) + \vec{b}(u), \frac{1}{2}\nabla(u_h - u_h^*)^2\right)\\ &= -\frac{1}{2}((\vec{b}(u_H) - \vec{b}(u)), \nabla(u_h - u_h^*)^2) - \frac{1}{2}(\nabla\cdot\vec{b}(u), (u_h - u_h^*)^2)\\ &:= B_{31} + B_{32},\end{aligned} \tag{55}$$

where,

$$B_{31} \le C\|u_H - u\|_0\|u_h - u_h^*\|_0^2 \le CH^{\frac{3}{2}}\|u_h - u_h^*\|_0^2. \tag{56}$$

So

$$\begin{aligned}A_2 + B_{32} &= (a(u_H)\nabla(u_h - u_h^*), \nabla(u_h - u_h^*)) + \frac{1}{2}(\nabla\cdot\vec{b}(u), (u_h - u_h^*)^2)\\ &\le |A_1| + |B_1| + |B_2| + |B_{31}| + |B_4|\\ &\le C\left(h + H^{\frac{3}{2}}\right)\|\nabla(u_h - u_h^*)\|_0.\end{aligned} \tag{57}$$

Table 1
Errors and computational time of the combined finite element-upwind volume method.

h	L^2-norm	Ratio	H^1-norm	Ratio	Computational time (s)
1/8	3.6956×10^{-2}	2.027	3.2896×10^{-1}	1.910	1
1/16	1.8233×10^{-2}	1.959	1.7220×10^{-1}	1.584	2
1/32	9.3103×10^{-3}	1.970	1.0870×10^{-1}	1.527	4
1/64	4.7249×10^{-3}	1.982	7.1177×10^{-2}	1.496	13
1/128	2.3845×10^{-3}	1.988	4.7582×10^{-2}	1.412	196

Table 2
Errors and computational time of the streamline diffusion finite element method.

h	L^2-norm	Ratio	H^1-norm	Ratio	Computational time (s)	$\delta(h)$
1/8	6.2723×10^{-2}	2.804	8.5747×10^{-1}	1.889	2	1.5×10^{-6}
1/16	2.2366×10^{-2}	2.231	4.5388×10^{-1}	1.855	2	5×10^{-6}
1/32	1.0027×10^{-2}	2.763	2.4462×10^{-1}	1.821	4	2×10^{-6}
1/64	3.6286×10^{-3}	2.859	1.3435×10^{-1}	2.159	19	3.5×10^{-7}
1/128	1.2692×10^{-3}		6.2215×10^{-2}		5739	1.1×10^{-7}

Table 3
Errors and computational time of the two-grid method.

h	H	L^2-norm	Ratio	H^1-norm	Ratio	Computational time (s)
1/8	1/4	1.5438×10^{-1}	8.348	3.7435	4.433	2
1/64	1/16	1.8493×10^{-2}	8.221	8.4453×10^{-1}	8.784	9
1/516	1/64	2.2496×10^{-3}		9.6140×10^{-2}		118

Since $\nabla \cdot \vec{b}(u) > 0$,

$$\|\nabla(u_h - u_h^*)\|_0 \le C\left(h + H^{\frac{3}{2}}\right), \tag{58}$$

then

$$\|u_h - u_h^*\|_1 \le C\left(h + H^{\frac{3}{2}}\right). \tag{59}$$

From Section 2, we have

$$\|u - u_h\|_1 \le Ch, \tag{60}$$

then

$$\|u - u_h^*\|_1 \le \|u - u_h\|_1 + \|u_h - u_h^*\|_1 \le C\left(h + H^{\frac{3}{2}}\right). \quad \square \tag{61}$$

4. Numerical example

In this section, we solve a model problem to demonstrate the efficiency and accuracy of our proposed numerical methods. Let $\Omega = (0, 1) \times (0, 1)$, the right-hand side term $f(u)$ is chosen such that the exact solution u of (1) is given by

$$u = (x - x^2)(y - y^2)e^{x+y}, \tag{62}$$

where $a(u) = u + 1$, $\vec{b}(u) = 10^5((x^2 - 3x + 3)(1 - 2y)e^{x+y}, u)$.

Our main interest is to verify the performance of the two-grid combined finite element-upwind volume method shown in Algorithm 3.1. Choose the space step H and then obtain the coarse grids. Let $h = H^{\frac{3}{2}}$ and then obtain the fine grids. In order to demonstrate the efficiency of the two-grid combined finite element-upwind volume method, we compare this method with the one-grid combined finite element-upwind volume method shown in Section 2. Computational results are shown in Tables 1–3. The computer platform we employ to implement the numerical simulation is Dell Dimension 9200 equipped with 2.13 GHz Intel Core CPU and 3.00 GB RAM.

In Table 1, with the standard combined finite element-upwind volume method, the convergence errors $\|u - u_h\|_0$ (labeled "L^2-norm") and $\|u - u_h\|_1$ (labeled "H^1-norm") are reduced by a factor of around 2 (indicated by the values in columns "Ratio") for each doubling of resolution of h, which is coincide with Theorem 2.1.

In Table 2, with the streamline diffusion finite element method, the convergence error $\|u - u_h\|_0$ (labeled "L^2-norm") is reduced by a factor of around 2.828 (approximately equal to $2^{\frac{3}{2}}$) (indicated by the values in columns "Ratio") for each

doubling of resolution of h. However, with the same accuracy, the combined finite element-upwind volume method is effective for saving more computational time in contrast with the streamline diffusion finite element method.

In Table 3, the convergence errors $\|u - u_h\|_0$ and $\|u - u_h\|_1$ are reduced by a factor of around 8 for each quadrupling of resolution of H and octupling of resolution of h, which is coincide with Theorem 3.1. Also, the two-grid combined finite element-upwind volume method spends less time than the one-grid combined finite element-upwind volume method on the comparable fine grid, that is to say, the two-grid algorithm is effective for saving a large amount of computational time and still keeps a good accuracy.

5. Conclusion

We studied a combined finite element-upwind finite volume method for a fully nonlinear convection-dominated diffusion reaction equation in this paper, derived its priori optimal error estimate in H^1-norm and sub-optimal error estimate in L^2-norm for piecewise linear finite element combining with the first order upwind finite volume scheme. To reduce the computational cost while keeping the approximation accuracy unchanged, we also developed a type of two-grid method for the present model with the combined finite element-upwind finite volume method on the fine grid T_h and the streamline diffusion finite element scheme on the coarse grid T_H. A priori error estimate of such designed two-grid method in H^1-norm is proved to be $O(h + H^{\frac{3}{2}})$. The results of numerical experiments demonstrated the efficiency and accuracy of our presented numerical method.

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